

Discovering JEPA Models

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1 Introduction

I started out this class by introducing myself as an Anthropology + CS/Data Science person who loves understanding the boundaries of technology and people. While I do LLM and foundational AI research across my labs, I always find my social science brain whispering that a lot is missing from these models. One of the key things that seems very clear is that natural language alone or predictive models are not encompassing of how people think and operate. There must be things beyond that that make us human and make us great at the tasks we execute.

That's where I find the new branch of AI startups, like Yann LeCun's AMI Labs and Fei-Fei Li's World Labs, really exciting as they are moving into the 3D data collection and models, also coined World Models. While I don't see it as a path towards equalizing machines and humans' abilities, it does provide a very interesting non-language based approach to developing AI. Throughout this project, I tried to understand the literature behind Yann LeCun's approach to world-building and then experimented with coding aspects of it with AI to feel their impact on my fingertips.

2 Shared Vision

Based on the interviews and position papers of Fei-Fei Li and Yann LeCun, both share a fundamental hope that world models will bridge the gap between current narrow AI (like LLMs) and human-level Autonomous Machine Intelligence. While they agree that language alone is insufficient for intelligence, their specific visions for achieving this differ in emphasis.

Fei-Fei Li’s *The Worlds I See* is one of my favorite books, so I am biased when I say that I love her approach to world-building: Her hope is to create AI that can reason with the precision of a designer, create with a storyteller’s imagination, and interact with the environment like a first responder. She argues that while text was the interface for LLMs, 3D representations are the equivalent medium for describing and interacting with the physical world (Y Combinator, 2025). The values and tools she is building out in World Labs are based on Human-Centered AI and Embodied Intelligence with the goal of building tools that are significantly better than LLMs (as they lack experience in the physical world) in assisting healthcare, care-givers, and robotics (Y Combinator, 2025).

Yann LeCun’s has been working on the “Billion-Dollar Bet” that extends predictive architecture beyond language as well. LeCun’s vision is focused on Autonomous Machine Intelligence that can learn as efficiently as humans—for example, a teenager learning to drive in just 20 hours (Welch Labs, 2026). Unlike Li’s focus on generating 3D worlds, LeCun argues that generative models (predicting pixels) are doomed to fail because they spend too much effort on irrelevant details (like leaves moving in the wind). JEPA’s goal is to ignore these details and focus on semantic content. While LeCun’s approach reads much less human-centric than Li’s, it’s much easier to replicate it. He is right that data collection for World Labs’ approach is impossibly expensive and very few academic articles have come out of World Labs so far (Welch Labs, 2026). LeCun’s approach is more public, tested, and easier to play around with for this project.

Table 1: Similarities and Differences in Approaches

Feature	Fei-Fei Li	Yann LeCun
Primary Goal	Spatial Intelligence and 3D world creation	Autonomous Intelligence and efficient learning
Core Architecture	Marble (Generative 3D scenes)	JEPA (Non-generative predictive world model)
Approach to Data	Use 2D video/images to extract 3D/4D structure	Self-supervised learning to capture dependencies in representation space
Major Application	Creativity, storytelling, and human-centered care	Robotics , long-term planning, and skills acquisition

3 Understanding JEPPA

The Joint Embedding Predictive Architecture (JEPA) is a non-generative framework for building predictive world models, proposed by Yann LeCun as a path towards autonomous machine intelligence (LeCun et al., 2022). Unlike traditional Large Language Models (LLMs) or Masked Autoencoders (MAE), which are generative and predict raw data (like pixels or text tokens), JEPA performs predictions in a highly abstract representation space (LeCun et al., 2022). The Core mechanics of a JEPA model rely on three main components: a **context encoder**, a **target encoder**, and a **predictor** (Assran et al., 2023; Maes et al., 2026).

- **Encoders:** Both the input (context) x and the target y are passed through encoders to produce latent embeddings, s_x and s_y . This allows the model to eliminate irrelevant, unpredictable details, such as the exact movement of leaves in the wind or the texture of a sidewalk, focusing instead on semantic content.
- **The Predictor:** This module is trained to predict the representation s_y from s_x . To handle the inherent uncertainty of the real world, e.g., a car approaching a fork in the road, the predictor can use a latent variable z . By varying z , the model can represent a set of multiple plausible future states.
- **Non-Generative Nature:** Because it does not reconstruct pixels, JEPA avoids the “blur-

riness” common in generative video models where compounding errors lead to a loss of structure over time.

Challenges that arise with JEPA

A significant hurdle for joint-embedding architectures is representation collapse, where the encoders learn to output a constant, uninformative vector for every input to “cheat” the prediction task (LeCun et al., 2022). Recent papers show that JEPAs have a distinct implicit bias compared to reconstruction-based models (Littwin et al., 2024):

- **Asymmetric Architectures:** Systems like I-JEPA use an Exponential Moving Average (EMA) of the context encoder’s weights to update the target encoder, coupled with a StopGrad operator to prevent gradient flow through the target branch (Littwin et al., 2024).
- **Regularization:** Methods like VICReg or SIGReg explicitly force the embeddings to maintain high information content or follow a specific distribution (like an isotropic Gaussian), ensuring feature diversity (LeCun et al., 2022).

While MAEs prioritize learning any feature with high variance (including noise), JEPAs are biased toward learning “influential features” with high regression coefficients that are truly predictive of future states (Littwin et al., 2024). As we know well from LLM research, debiasing the datasets is very hard, and when working with data beyond language, the tricky dimensions are only amplified. However, with JEPA being a newer approach that hasn’t been pushed into public usage yet, a more critical approach to data collection, annotation, and transparency should be key to work through.

4 Coding Segment

While I spent a majority of this project reading papers and listening to Yann LeCun’s lectures to understand the approach, we also discussed trying out this “toy” and attempting to play with the

nodes in the approach.

An example for experiment

The example I am drawing on comes from the LeWorldModel paper (Maes et al., 2026): a 2D grid world where a dot moves around. In the coding segment found on my GIT Hub, I decided to work through the pipeline that is mappable to the paper (Maes et al., 2026):

- Step 1 → Section 3.1 of the paper (Training the Latent World Model): dot moves in a 2D plane following simple physics (velocity + walls). This mirrors the Two-Room and PushT environments. We generate synthetic trajectories of (observation, action) pairs offline. Observation = 16x16 pixel frame with a dot. Action = 2D velocity command.
- Step 2 → Figure 1 + Section 3.1 (Encoder + Predictor + SIGReg): Train the miniature LeWM: ViT → replaced by small CNN encoder, transformer predictor → replaced by 2-layer MLP. Loss = MSE prediction + λ ·SIGReg. Show collapse happens without SIGReg, doesn't happen with it.
- Step 3 → Section 5.1 + Table 1 (Physical Latent Probing): Train a linear probe on frozen latents to recover the dot's (x,y) position. Mirrors exactly Table 1's "Agent Location" row. Show r^2 before and after training, and with/without SIGReg.
- Step 4 → Figure 10 + Section 5.2 (Violation-of-Expectation): Run trained model on normal trajectories vs. teleportation trajectories. Plot MSE surprise spike. Directly replicates Figure 10's structure (Maes et al., 2026).
- Step 5 → Section 4 + Figure 6 (Hyperparameter sweep): Vary λ (SIGReg weight), embedding dimension, and predictor size. Show performance curves. Mirrors Figure 15 and 16 ablations (Maes et al., 2026).

At the end, there is a pipeline connector to unite the above steps.

Results of running the model

As we trained the 32x32 pixel environment where a dot bounces around a box, the code generated 500 training trajectories and 40 test trajectories (20 normal, 20 with teleportation at step 25). We trained two versions of the model. The one with SIGReg ($\lambda=0.1$) shows a loss around 0.00084 that gradually decreases. The one without SIGReg ($\lambda=0$) hits exactly 0.00000. That zero loss is the collapse the paper warns about in Section 3.1: the encoder learned to map every frame to the same constant vector, which trivially satisfies the prediction objective because predicting a constant is easy (Maes et al., 2026). The regularized model maintains a nonzero loss because it is forced to keep embeddings spread out.

Stage 3 successfully mirrors Table 1 of the paper (Maes et al., 2026). The model tried to predict the dot’s actual (x,y) position from the learned latent vectors using a simple linear regression. The SIGReg model got a weak but real correlation ($r \approx 0.15$). The no-SIGReg model produced NaN because its encoder output a constant vector for every input, making correlation mathematically undefined. The probing numbers are weak compared to the paper’s $r=0.97$, which is expected given I trained for 50 epochs on a tiny CNN versus their 200k steps on a ViT, but the directional result is correct. This indicates one of the most significant issues with replicating JEPA, especially without a GPU: JEPA requires immense amount of data and compute. It makes sense that Yann LeCun is actively seeking compute funding and resources; without that it’s impossible to even show the impact and direction in which JEPA can lead (Welch Labs, 2026).

Mirroring Figure 10 of the paper (Maes et al., 2026), I got a surprise ratio of 3.74 at step 25. That means that when the dot teleports, the model’s prediction error spikes to 3.74 times its normal baseline. That is the violation-of-expectation result: the model learned enough about normal dot physics that an impossible jump registers as anomalous. This is the most convincing result in the entire run and didn’t require too many runs to solidify.

Finally, the R^2 values are all negative from the runs, which means the linear probe is doing worse than just predicting the mean. This points to a training depth issue: 50 epochs on a small model is not enough for the latent space to become well-organized for linear readout. However, the

trend across λ is meaningful: R^2 improves monotonically as λ increases from 0 to 0.5, which exactly matches Figure 16 of the paper. More regularization produces more structured, linearly-readable representations.

Table 2: Results Summary Table

Model	Probe MSE	Probe r	Surprise Ratio
With SIGReg ($\lambda = 0.1$)	47.6134	0.1548	3.74
Without SIGReg ($\lambda = 0$)	48.8204	nan	N/A

The figures that are attached in the data folder on GIT show the model replicated four distinct claims from the paper on a toy system in a few hours of CPU training. Collapse is real and visible. SIGReg prevents it. Physical structure is weakly but genuinely encoded when collapse is prevented. The model detects physically impossible events with a 3.74x surprise spike. The quantitative gap from the paper’s results is entirely explained by training scale, and the qualitative story is intact across every single figure.

AI Usage

This project made substantial use of Claude Code (Anthropic), an AI coding agent, to implement the full LeWM-mini codebase. The AI was used as a primary implementation partner: the agent wrote all Python source files from scratch, verified them with smoke tests. The debugging for issues that came up was done collaboratively. My tasks were primarily the lit review and absorption of how the approach works. I gave AI detailed specification of what to build and which paper sections to map each feature to

For me, using agentic coding wasn’t new at all. I use it all the time in my labs and for person project. However, what was the most significant win for me is that AI coding allowed me to replicate something from scratch that even a year ago required an advance degree and long time spent on the project. At the same time, this project allowed me to see why huge AI companies still hire teams of software engineers: It takes effort and time to understand what you are trying to create and to map part of your imagination with tools for its implementation. Finally, the big AI

labs are still ahead of an individual coder not only because of research capabilities, and not even because of compute – data is the biggest missing variable that large companies have effort, people, and tools to collect on an unimaginable scale. I really hope that world-building efforts will soon show cool products and wins as they are clearly building out the scope of their datasets.

5 What is still missing in this approach

While JEPA is a promising framework for world modeling, sources identify several critical gaps and remaining challenges:

- **The “Mysterious” Configurator:** In the proposed cognitive architecture, the configurator is intended to be the executive controller that sets goals and subgoals for the agent (LeCun et al., 2022). However, it is currently the “most mysterious” module, and there is no specified mechanism for how it learns to decompose complex tasks into manageable sub-objectives (LeCun et al., 2022).
- **Lack of Grounding Critiques:** Some researchers argue that by abandoning pixel-level reconstruction, JEPAs lose mathematical controllability (Xing et al., 2025). Without a decoder to “ground” representations in observable data, it can be difficult to diagnose if the model is learning meaningful dynamics or simply satisfying its regularizers (Xing et al., 2025).
- **Short Horizons and Hierarchy:** Current practical implementations like V-JEPA are often restricted to short temporal horizons (e.g., predicting gaps in a few seconds of video) Bandaru, 2024. While the Hierarchical JEPA (H-JEPA) is proposed to handle long-term planning across multiple time scales and levels of abstraction, building and training such a hierarchy from complex video remains an open question (LeCun et al., 2022).
- **Data and Interaction Dependencies:** Existing models heavily rely on offline datasets with high interaction coverage, which are expensive to collect (Maes et al., 2026). Additionally, most JEPA models still require explicit action labels to predict future states; a more

autonomous path would involve learning action representations through inverse dynamics modeling without human-provided labels (Maes et al., 2026).

6 Looking ahead

Through this project, I replicated the LeWorld paper’s qualitative story correctly. Collapse happens without SIGReg. SIGReg prevents it. The surprise spike is real and substantial. The λ trend goes in the right direction. What I wasn’t able to replicate fully is the quantitative performance, because the paper’s probing r values near 0.99 come from much longer training on a stronger architecture. If I wanted to push the probe r closer to the paper, the main levers are training longer (more epochs), increasing model capacity slightly, and possibly adding a learning rate schedule.

Finally, looking ahead at the usage of this technology, I hope Dr Li’s human-centric approach prevails while LeCun’s code transparency and reproducibility becomes widely-experimented. I was caught off-guard with the dissonance between Li’s beautiful descriptions of what non-language based models can do and LeCun’s “easy” explanations of how his technology works while the implementation is so math-y and complicated. Of course, I understood the limitations without access to a GPU but I never thought of that beautiful 3D models I see in World Labs’ advertisements all happen through good-old math :) Coming back to the AI disclosure and usage, that all reminds me that we need people who can dream up and imagine these futures and a lot of hands working to sift through the math before more people use and believe in the non-text-based models and intelligence. Maybe the difficulty of this project makes me happy as there is still so much for human intelligence to prevail and excel in!

References

- Assran, M., Duval, Q., Misra, I., Bojanowski, P., Vincent, P., Rabbat, M., LeCun, Y., & Ballas, N. (2023, April). Self-supervised learning from images with a joint-embedding predictive architecture. <https://arxiv.org/abs/2301.08243>
- Bandaru, R. (2024). Deep dive into yann lecun’s jepa. <https://rohitbandaru.github.io/blog/JEPA-Deep-Dive/>
- LeCun, Y., et al. (2022). A path towards autonomous machine intelligence version 0.9. 2, 2022-06-27. *Open Review*, 62(1), 1–62.
- Littwin, E., Saremi, O., Advani, M., Thilak, V., Nakkiran, P., Huang, C., & Susskind, J. (2024). How jepa avoids noisy features: The implicit bias of deep linear self distillation networks. *NeurIPS*. <https://arxiv.org/abs/2407.03475>
- Maes, L., Le Lidec, Q., Scieur, D., LeCun, Y., & Balestrieri, R. (2026, March). Leworldmodel: Stable end-to-end joint-embedding predictive architecture from pixels. <https://arxiv.org/abs/2603.19312>
- Welch Labs. (2026, May). Yann lecun’s \$1b bet against llms [Accessed 13 May 2026]. <https://www.youtube.com/watch?v=kYkIdXwW2AE>
- Xing, E., Deng, M., Hou, J., & Hu, Z. (2025, July). Critiques of world models. <https://doi.org/10.48550/arXiv.2507.05169>
- Y Combinator. (2025, July). Fei-Fei Li: Spatial Intelligence is the Next Frontier in AI [Fireside with Dr. Fei-Fei Li recorded on June 16, 2025 at AI Startup School in San Francisco]. <https://www.youtube.com/>